-RESERCH LOG BOOK-

KARUNARATHNE D M S G IT22263476
It22263476@my.sliit.lk

PEOPLE'S BANK

Date: _____

Reserch project With Log Activity

projectID → R26_DS_011

StudentID → IT22063476

Name → D. M. S. C. Karunarathna

Individual Compovent → Low Bandwikth / Text SMS Emergency Intelligence.

2025-12-03

Meet Our Supervisor and discuss about ~~Dew~~ topics to do Our reserch.

2025-12-10

Finalized Our reserch topic named 'AI based emergency Call and SMS triage system for disaster response in Sri Lanka Low resource Local Languages' as a Supervisor

2025-12-15

Meet Our reserch Supervisor and discussed Our Components with Other Members.

2025-12-20

Finalized reserch topics with Components

2025-12-21

Register Our reserch group and topic with Relevant Components.

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Date:

Student progressing well in reserch project.

Yes/No

Signature:- Identified the general reserch domain and discussed ~~the real - world~~ important



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Date: _____

Project ID - R-26-DS-011

Student ID - IT22265476

Name - D.M.S.G. Ranmorathna

Individual Compant - Low Bandwidth / Text SMS Emergency Intelligence.

26/01/01 : Identified the general reserch domain and discussed the real important of an AI-based emergency calls and sms triage system for disaster response in Sri Lanka, espedally for lower-resource Local Languages Such as Sinhala Tamil

26/01/04 :

Researched how emergency Communication System coork and focused on the challages In Processing multilingual emergency Calls and SMS Messages in Sinhala, Tamil, mixed Language Input.

26/01/07 :

Collected related reserch papers and reviewed previous studies Connected to speech text Emergency Massage analysis.

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Date:

2026/01/10:
Visited DMC for the first time discussed Our reserch topic. we talk about the Importance of this reserch to DMC and discussed possible ways TO collect data for Our study

2026-01-10:
prepared my Part of TAF document and focused on how the system analyze emergency

26-01-18:
Submited the TAF Document.

Student progressing well in reserch project
Yes/No

Signature:



PEOPLE'S BANK

Date: _____

ProjectID :- 2-26-DS-011

StudentID : IT22265476

Name : DMSG Kawrathna

Individual Component : Low Bandwidth / Text SMS emergency Intelligence

20/02/04 :- Started to identify what data should be Collected for my function and discussed the main inputs needed for the emergency

26/02/04 :-

Attended a meeting and discussed the function with Industry related experts Gained useful ideas about emergency Communication - Language challenges and possible for inputs.

26-02-12 :

prepare the external data collection request form and submited it to CDAP for data Collection process.

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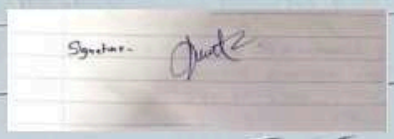
Date:

2026-02-14: Arrange a metting with group members and discussed each members user inputs and Outputs to keep the overall system Connected properly

26/02/26: Arrange a metting with group members and Supervisor and presented our inputs outputs After the discussion, made several Changes to improve the function.

Student progressing well in research proj

Yes/No.



Signature: ------

PEOPLES BANK

Date:

PS-26-DS-011

Project ID: ~~IT2226476~~

Student ID: IT22263476

Name: DMSC Kawarathna

Individual Component: - Low Bandwidth/ Text SMS Emergency Intelligence

2026/03-01: - Started preparing a research proposal presentation and organize the idea for the research topic

26-03-05: - Identified the structure of the individual report presentation selected of the content related to my component.

26-03-07: -

Collected ~~and identified~~ the external data collection Letter from CDAP visited DMC & hand over the Letter.

26-03-22

Arrange a meeting with group members & supervisor to discuss each members's of report and give feedback for improvement.

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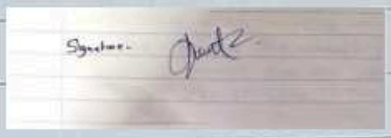
Date:

26 - 03 - 26

presented the reserch proposal Presentation finalized the reserch proposal presentation and Obtained supervisor signature for the proposal

26/03/26 - presented the reserch proposal

Student progressing well in Research project Yes / No

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PEOPLE'S BANK

Date:

project ID :- R-26-DS-011

Student ID - IT22263476

Name :- D MSG Kanishka Ashan

Individual Component - Low Bandwidth / Text SMS emergency Intelligence

2026-04/01-

Arrange a meeting with Supervisor and group members to discuss the project.

26/04/09:

Serched for the suitable models to use in My research Component and gained additional knowledge about AI models related functions

26-04/09:- Started collecting data for the Research.

Student proceeding well in Research project year/m

Signature :



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Date:

Project ID :- R26-DS-011

Student ID - IT22262476

Name - D M S G Karunarathna

Individual Component - Loco Bandwith Text/sms Intelligent System.

2026-04-10 - Collected the Data Needed for my function

2026-04-11 - Sort the Dataset and did more reserch about the Dataset

2026-04-15 - Completed the data Sorting and began the proio of identifing Convertional Cues

2026-04-20 - Had a Team meding to discuss The project plan and decide task Amoug members

2026-04-24 - sorted deducing model building based on prosodic and Convootional Cases

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Date

2026-04-26 Had a Team meeting to discuss The progress as a Team

2026-04-27 Had a discussion with DMC over The phone on the prices of Data Collection.

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Date:

Project ID - R26-DS-011

Student ID - IT22263476

Name - D M S A Karunarathna

Individual Component - Low Bandwidth Text/SMS Intelligent System

01-05-2026 - Start a project and identified the problem to solve (emergency/SMS assist)

~~04-05-2026~~

04-05-2026 - Researched existing disaster alert and disaster response system

05-05-2026 - Defined project objectives and its main system feature

06-05-2026 - Planned the system workflow and designed the basic Architecture

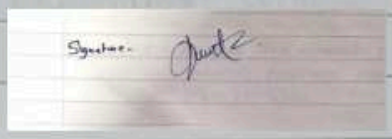
07-05-2026 - Collected. Sample emergency sms emergency and disaster related text data

08-05-2026 - - cleaned and Organized the dataset for Processing

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Date: _____

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Date:

11-05-2026 - Studied NLP techniques for text processing classification

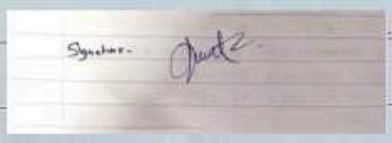
12-05-2026 - Set up & develop environment & installed required tools and Libries

13-05-2026 - Started implementing text processing and Normalizing

14-05-2026 - Tested different machine Learning models for sms classification

15-05-2026 - Evaluated Model results and selecting the best model for further development

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Date:

18 -5-26 - Evaluated all trained models using performance metrics and identified best model

19 -5-26 - Analyse strength & weakness of the selected model & documet the evaluation

20 -5-26 - Review model predictions & validate its performance again test dataset

21 -5-26 - optimized the selected model by fine tuning key parameters

22 - 5 - 26 - Conduct additional testing to ensure the model perform consistency across scenarios.

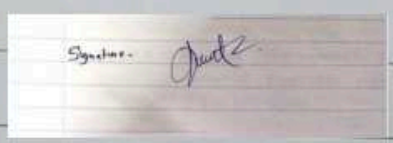
25 - 5 - 26 - prepare the selected model for integration into development

26 -5 - 26 - Investigated model errors and identified potential cases

27 -5 - 26 - Documented model Architecture evaluation outcomes

29 -5 - 26 - finalized the selected model & established the roadmap

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Date:

01-01-2026 Integrated multi Agent routing System with Language detection pipline.

02-6-2026 Debugged incident type classification, Fixed edge case for single letter inputs

03-6-2026 Implemented priority scoring Algorithm with 6-level triage system

04-6-2026 - Build phonetonic location match for 57,600+ Srilankan place name

05-6-2026 Created Gen2 Slang dictionary with 399 term and context aware translation Logic

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Date: _____

08-06-2026 Integrated Location Confirmation multi agent flow's tested accuracy with 100 test cases.

9-6-2026 Fixed timezone handeling for dispatch queue timestamps

10-6-26 - Implemented responsive Chat UI with progress bar component

11-6-26 - Built message formatting engine with dynamic stylying based on priority level

12-6-26 - Tested Conversation flow end-to end with all 4 field collection steps

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Date:

15-6-21 - Added support for 6 languages Varisanta Comglish, singlish, Tanglish, sinhala, Tamil, mixed

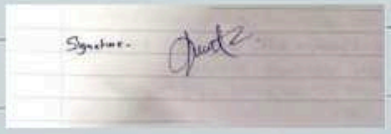
16-6-21 - Debugged languages detection confidence scoring; improve accuracy to 95%.

17-6-21 - Tested Integrated twillio sms webhook handler for incoming message routing

18-6-21 Build case ID generation and dispatch queue management system.

19-6-21 - Create priority banner display component; tested across mobile / desktop

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Date:

6-7-26 Debugged Location typo handling & improve phonetic matching 85% accuracy.

7-7-26 Enhanced Gen Slang detection with context awareness Validation

8-7-26 Fixed priority Calculation edge cases (Fire + people = Eco Confirmed).

9-7-26 Implemented message styling for all priority levels with color coding

10-7-26 Tested response time performance & achieved <2s average.

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Date:

13-7-26 Added support for ambiguous location suggestions (3+ matches display)

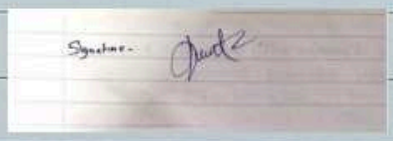
14-7-26 Build people count validation with 'don't know' fallback

15-7-26 Implemented resource requirement categorization (ambulance, fire truck, police)

16-7-26 Created comprehensive error handling for malformed input message

17-7-26 Tested special character handling and HTML injection prevention

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Date: _____

20-7-26 Debugged message tag system (connection, Pensionne, update)

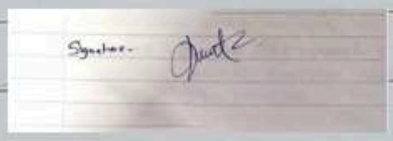
21-7-26 Enhanced progress bar animation & step completion detection

22-7-26 fixed database connection pooling for concurrent users handling

23-7-26 Implemant message peristence and retrieval from log database

24-7-26 Tested load handling with 50+ simulated concurrent Conversations.

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Date:

29-7-26 Added Api endpoint and logging for all transaction

28-7-26 - Implemented graceful degradation for missing Location date

29-7-26 - Create incident type auto location using fuzzy matching

30-7-26 Tested false positive rate for emergency detection (target: <1%)

31-7-26 performance profiling completed; optimized database queries.

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Date: _____

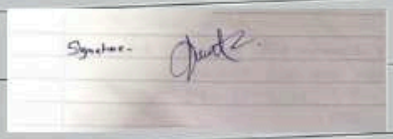
3-8-26 Implemented caching layer for frequently matched location (400+ cached)

4-8-26 Added multi-dialect Sinhala / Tamil script recognization

5-8-26 Build confidence scoring for language detection (threshold 85%)

6-8-26 Enhanced conversation state machine with rollback functionality

7-8-26 Tested memory usage under sustained load (100+ messge / minute)

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PEOPLE'S BANK

Date: _____

10-8-26 - Implemented rate limiting (10 msgs) with user ID tracking

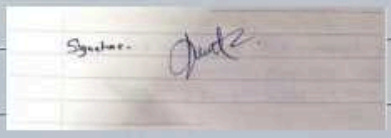
11-8-26 - Added detailed audit loging for all dispatch decision.

12-8-26 Fixed timeout issues in Coeation Confirmation flow (30s limit)

13-8-26 Integrated sms delivery Confirmations webhook handling

14-8-26 Tested sms delivery latency across multiple Twillio region

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PEOPLE'S BANK

Date: _____

17-8-26 Implemented fallback response templates for network failure.

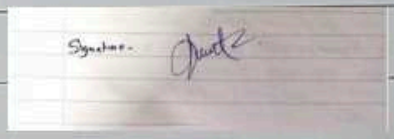
18-8-26 - Build dashboard for monitoring active Conversation (real-time)

19-8-26 added export functionality for dispatch queue (CSV → JSON format)

20-8-26 - Debugged character encoding issues for Tamil / Sinhala Text

21-8-26 Implemented full text search for case history Lookup.

Signature: -



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24-8-26 Enhanced prioritiy recalculation for field updates mid-Conversation

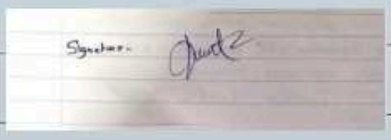
25-8-26 Created alert system for hight priority case (Echo, DELTA)

26-8-26 Tested mobile keyword handling (on-screen keyboard impact on UX)

27-8-26 Implemented location perCISPon Validation (min coordinates accuracy)

28-8-26 Added supervised learning feedback loop for slang dictionary update

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PEOPLE'S BANK

Date: _____

31-8-26 - Tested system stability after 100+ hour continuous run

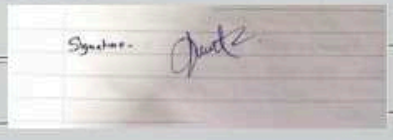
1-9-26 - Integrated vector embedding for semantic similarity in location matching

2-9-26 - optimized regex patterns for incident type detection (40% faster)

3-9-26 - Implemented context-aware follow up question generation

4-9-26 - Build multi language response scoring

signature:



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Date:

7-9-26 - Debugged async/await issues in concurrent message processing

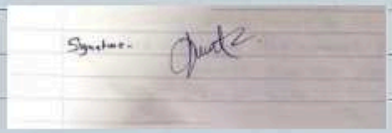
8-9-26 - create health check endpoints for system monitoring

9-9-26 - Implemented circuit breaker pattern for external API calls.

10-9-26 Enhanced error messages with actionable recovery suggestions

11-9-26 Tested disaster recovery with simulated database failure.

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PEOPLE'S BANK

Date: _____

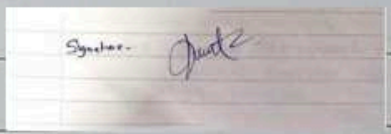
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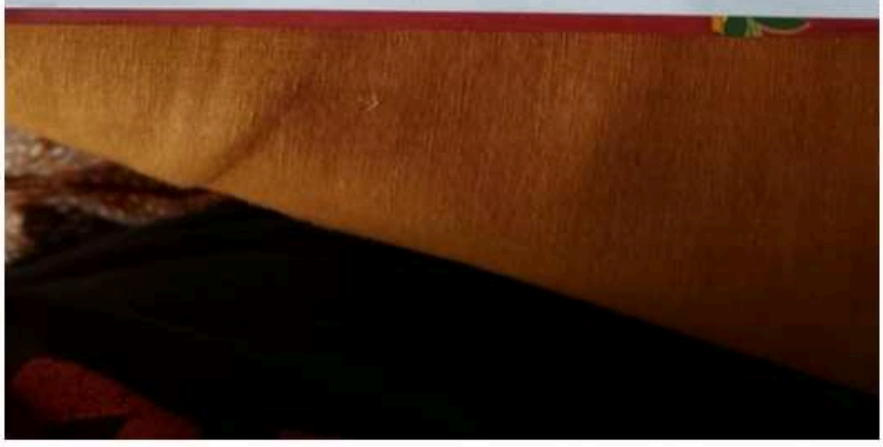
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PEOPLE'S BANK

Date: _____

14-9-26 Implemented Redis caching layer for location lookup (99.2% hit rate)

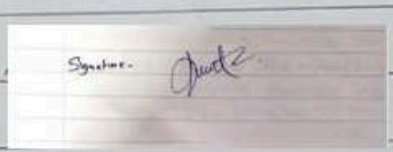
15-9-26 Build analytics dashboard showing case distribution and priority breakdown

16-9-26 - Created automated backup system for dispatch queue

17-9-26 - Enhanced slang detection with phrase-level matching

18-9-26 - Tested predictive typing suggestions for location field

Signature: _____



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Date: _____

21-9-26 Implemented voice input support (transaction + language detection)

22-9-26 - Build Compliance logging for GDPR / privacy regulations.

23-9-26 - Debugged unicode normalization issue in phonetic matching

24-9-26 - Enhanced notification system with SMS delivery ~~receipt~~ receipts.

25-9-26 Tested integration with emergency service API

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Date: _____

28-9-26 :- Implemented A/B testing framework for response

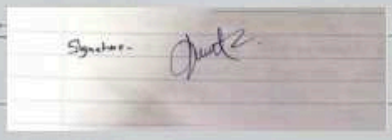
29-9-26 Build conversation quality scoring (1-10 scale).

30-9-26 optimized LLM API Calls; reduced token usage by 35%.

1-10-26 - created Comprehensive documentation for All APIs and function

2-10-26 - Build deployment script for Staging & production environment

signature :- ---



PEOPLE'S BANK

Date:

5-10-26- Implemented automated regression test suit (50+ test cases)

6-10-26- Enhanced error reporting with stack traces and context data
Tested failover mechanisms for redundant database replicas

7-10-26 -Build monitoring alerts for SLA violations (99.5%)
Debugged edge cases discovered in integration testing
enhanced multi-user conversation isolation & data privacy.

8-10-26- Implemented request signing for API authentication.

9-10-26- Created performance baseline metrics for regression detection

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Date:

12-10-26 - Tested localization for all date/time format across religion
Build emergency escalation protocol for critical failure scenarios.

13-10-26 - Implemented geo-fencing logic for location-based response routing.

14-10-26 - Enhanced security: SQL injection and XXS prevention verification
Build incident categorization model
Tested chatbot with accessibility features

15-10-26 - Implemented version control system for All API responses.

16-10-26 - Enhanced deployment pipeline with blue green deployment strategy.

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Date:

19-10-21 - Build Comprehensive User feedback Collection & analysis system

~~[illegible]~~ Final regression testing across all browser and platform

~~[illegible]~~ - prepare production release notes and deployment checklist.

20-10-21 - prepare production release notes and deployment checklist

Completed Security audit & penetration ~~23-10-21~~ - testing

21-10-21 - Final Verification All System operational ready for production deployment.

Signature:

